

SpaceX Falcon 9 Landing Prediction

Applied data collection, SQL analysis, interactive visualization, and classification modeling

GitHub repository: <https://github.com/cooper-710/Coursera-Capstone>

Executive Summary

- This project predicts Falcon 9 first-stage landing success using launch records, launch site data, payload mass, orbit, booster information, and engineered classification features.
- The workflow covered API collection, web scraping, wrangling, SQL analysis, exploratory visualization, Folium mapping, a Plotly Dash dashboard, and machine learning classification.
- EDA showed landing success improved over time and varied by launch site, orbit type, payload range, and booster version.
- Four classification models were tested. Logistic Regression, SVM, Decision Tree, and KNN all reached about 83.3% test accuracy on the held-out test set.
- The final model predicted successful landings well, but it was weaker at identifying failed landings.

Primary takeaway: SpaceX landing outcomes are strongly tied to launch maturity, launch context, and vehicle configuration, but failure prediction remains the harder task.

Introduction: Business Problem

SpaceX reduces launch cost by recovering and reusing the Falcon 9 first stage. Predicting whether the first stage will land successfully helps estimate launch economics, operational risk, and mission planning confidence.

1

Business question

Can launch data predict whether Falcon 9 first stage recovery will succeed?

2

Data science goal

Build a complete workflow from raw data collection to predictive classification.

3

Decision value

Identify launch patterns that increase or reduce landing success likelihood.

- Target variable: Class, where 1 indicates landing success and 0 indicates unsuccessful or no landing.
- Core features: payload mass, orbit, launch site, flight number, booster version, reuse indicators, and launch hardware variables.
- Final deliverable: PDF report plus GitHub repository containing notebooks, dashboard app, and screenshots.

Project Workflow

1

Collect

SpaceX API and web scraping

2

Wrangle

Clean, filter Falcon 9, build Class label

3

Explore

EDA charts and SQL summaries

4

Visualize

Folium maps and Dash dashboard

5

Model

Classification with tuned ML models

6

Report

GitHub, PowerPoint, and PDF submission

The project was built as a technical report, not just a high-level executive summary. Each section includes method, result, and interpretation.

GitHub Repository

Repository link

<https://github.com/cooper-710/Coursera-Capstone>

Folder / File	Purpose
notebooks/	All completed Jupyter notebooks for data collection, wrangling, EDA, SQL, maps, and machine learning
dashboard/	Plotly Dash application file
images/	Screenshots and chart outputs used in the final report
README.md	Project overview, methods, questions, and repository structure
requirements.txt	Python packages needed to rerun the analysis

The GitHub repository is included in the PDF report to satisfy the project deliverable requirement.

Data Collection Methodology

1

SpaceX API

Collected structured launch details including payloads, launch sites, booster data, landing outcomes, dates, and mission metadata.

2

Web scraping

Scraped Falcon 9 launch records from a historical launch table to supplement API data and practice HTML table extraction.

- API notebook output included normalized launch data and saved the cleaned records for downstream tasks.
- Web scraping notebook parsed HTML tables, cleaned inconsistent text, and exported a structured CSV.
- The combined workflow created a reproducible data source for wrangling, SQL, EDA, dashboarding, and prediction.

Notebook	Main deliverable
jupyter-labs-spacex-data-collection-api.ipynb	API dataset from SpaceX launch records
jupyter-labs-webscraping.ipynb	Scraped Falcon 9 launch table

Data Wrangling and Feature Engineering

- Filtered the dataset to Falcon 9 records only, excluding Falcon 1.
- Handled missing payload mass values by using the calculated mean payload mass.
- Converted raw landing outcomes into a binary Class target for machine learning.
- Created features for launch site, orbit, payload mass, flight count, grid fins, reused hardware, legs, landing pad, and booster block.
- Exported cleaned data for later EDA, dashboard, and modeling notebooks.

Wrangling output	Value
Target field	Class
Success label	1 = landed
Failure label	0 = did not land
Common launch sites	CCAFS SLC 40, KSC LC 39A, VAFB SLC 4E
High-count orbits	GTO, ISS, VLEO, PO

This step changed the project from raw historical launch records into a usable supervised classification dataset.

EDA and Visual Analytics Methodology

- Used pandas, seaborn, and matplotlib to compare landing outcome patterns across payload mass, flight number, launch site, orbit type, and year.
- Used SQL to answer structured questions from the launch table, including site lists, payload summaries, outcome counts, and date-filtered records.
- Used Folium for spatial analysis of launch sites and nearby geography.
- Used Plotly Dash for an interactive dashboard with dropdown, slider, pie chart, and scatter plot callbacks.

1

Static EDA

Scatter, bar, and line charts

2

SQL EDA

Relational queries and grouped summaries

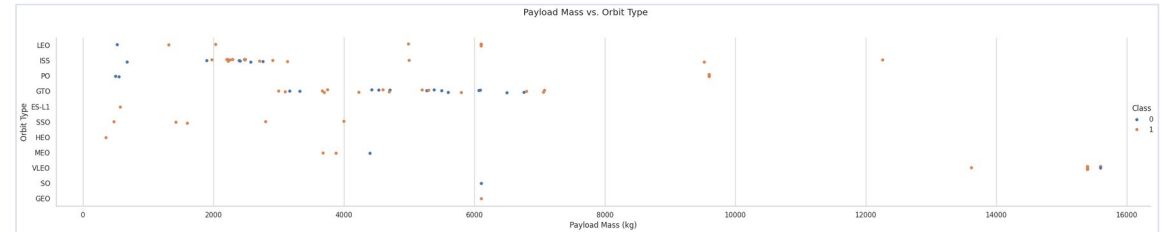
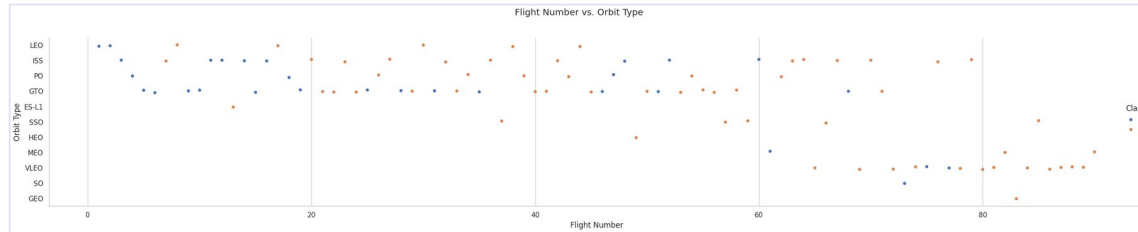
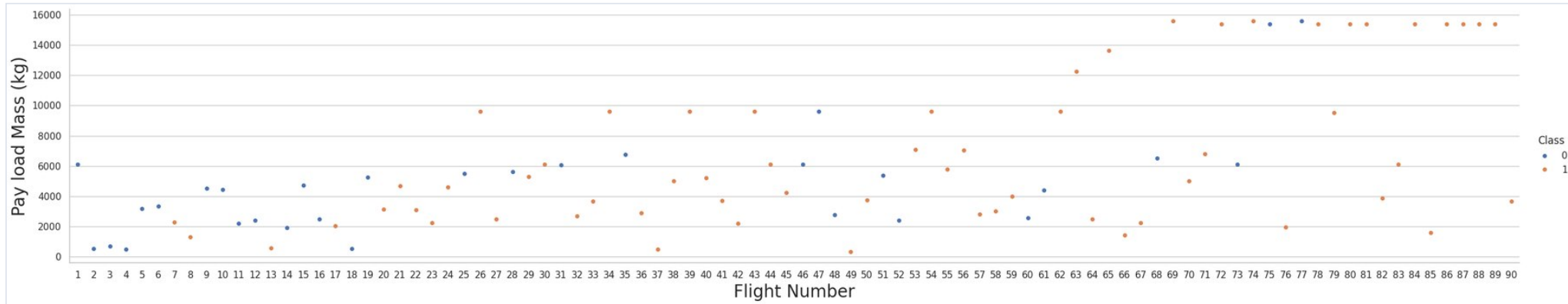
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Interactive visual analytics

Maps and dashboard filters

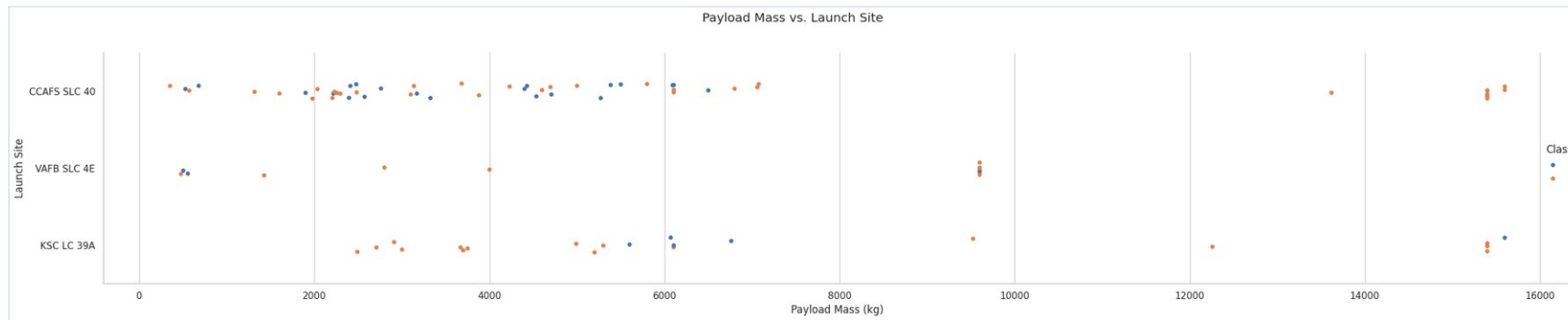
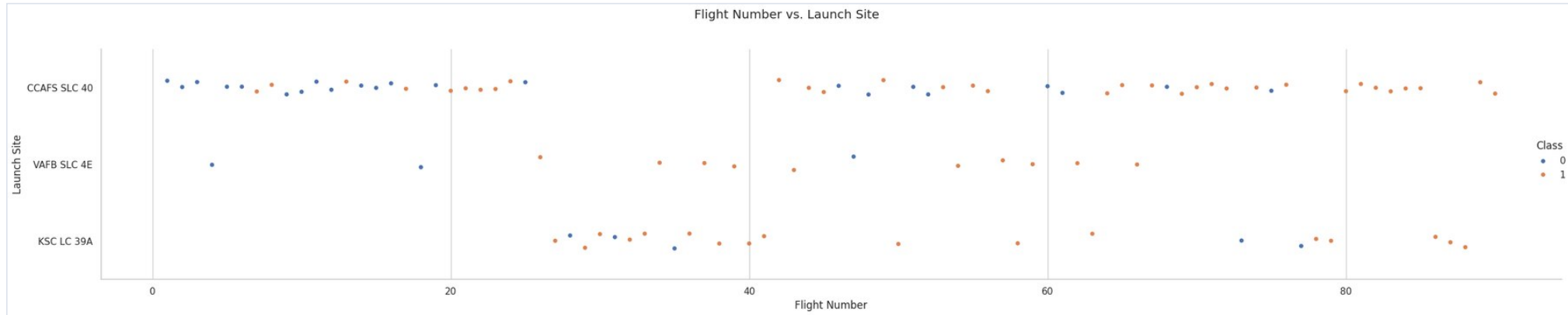
The analysis focused on both interpretability and repeatability, with each method producing outputs for the final report.

EDA Results: Flight Number and Payload Mass



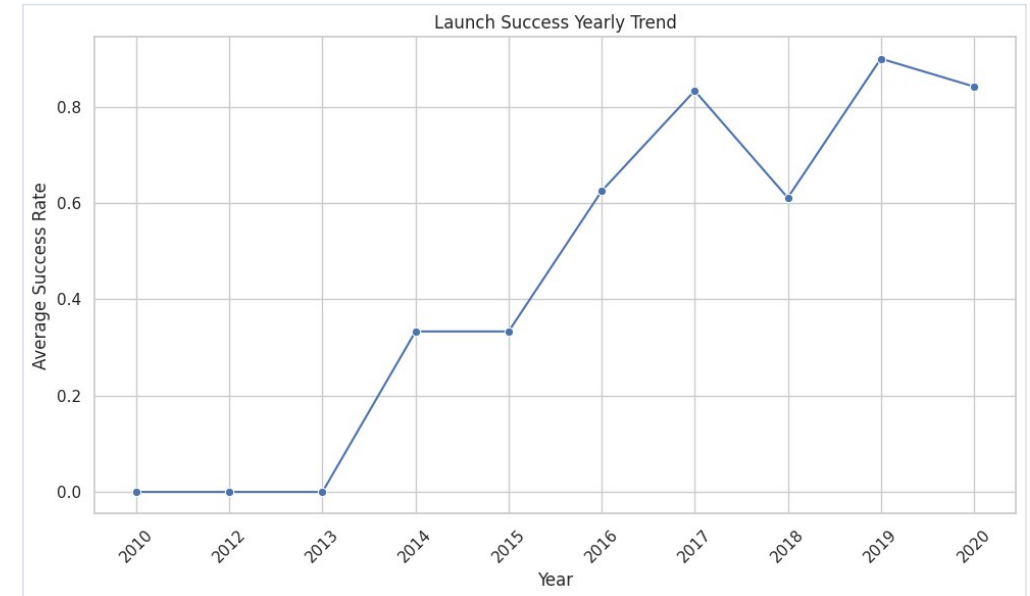
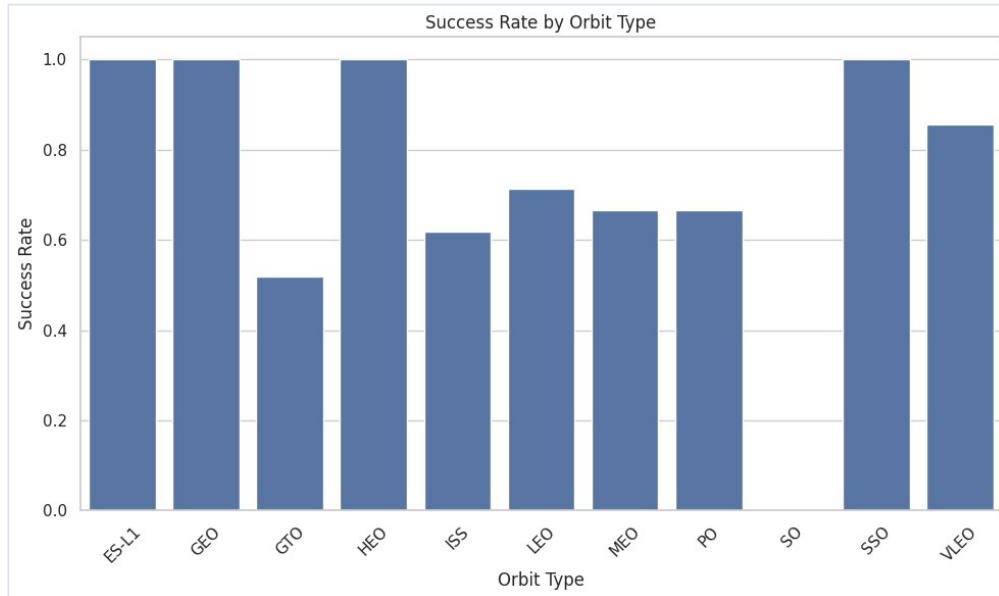
- Later flights show a higher concentration of successful outcomes.
- Payload mass alone does not fully explain landing success, but it interacts with launch site, orbit, and booster maturity.
- Flight number is a useful proxy for operational learning and program maturity.

EDA Results: Launch Site Patterns



Launch sites show different success patterns because they support different mission profiles, payload ranges, and orbital requirements.

EDA Results: Orbit Type and Yearly Trend



- Some orbit types had noticeably higher success rates than others.
- Average yearly success rate rose sharply after the early Falcon 9 years.
- The trend supports the idea that reuse and recovery capability improved as SpaceX gained launch experience.

SQL Methodology

Methodology

- Loaded the cleaned SpaceX CSV into a SQLite database.
- Created a working launch table with non-null dates.
- Used SQL to summarize payload mass, launch sites, booster versions, landing outcomes, and date-specific records.
- SQL results were used to validate the exploratory analysis and provide reproducible tabular findings.

SQL task category	Example query result
Launch site list	Four distinct launch sites found
Payload aggregation	NASA CRS payload total = 45,596 kg
Booster filtering	Average payload for F9 v1.1 = 2,928.4 kg
Date filtering	First successful ground pad landing = 2015-12-22
Outcome grouping	Mission outcomes and landing outcomes counted by category

SQL Results: Core Summary Queries

Question	SQL result
Distinct launch sites	CCAFS LC-40, VAFB SLC-4E, KSC LC-39A, CCAFS SLC-40
Total payload mass for NASA CRS	45,596 kg
Average payload mass for F9 v1.1	2,928.4 kg
First successful ground pad landing	2015-12-22

- The SQL results confirm that the dataset includes multiple launch facilities and mission profiles.
- Payload mass summaries give context for mission scale and customer-specific launch activity.
- The first ground pad recovery date provides a clear milestone in the recovery timeline.

SQL adds structure and auditability. The same questions can be rerun directly against the cleaned launch table.

SQL Results: Landing Outcome Patterns



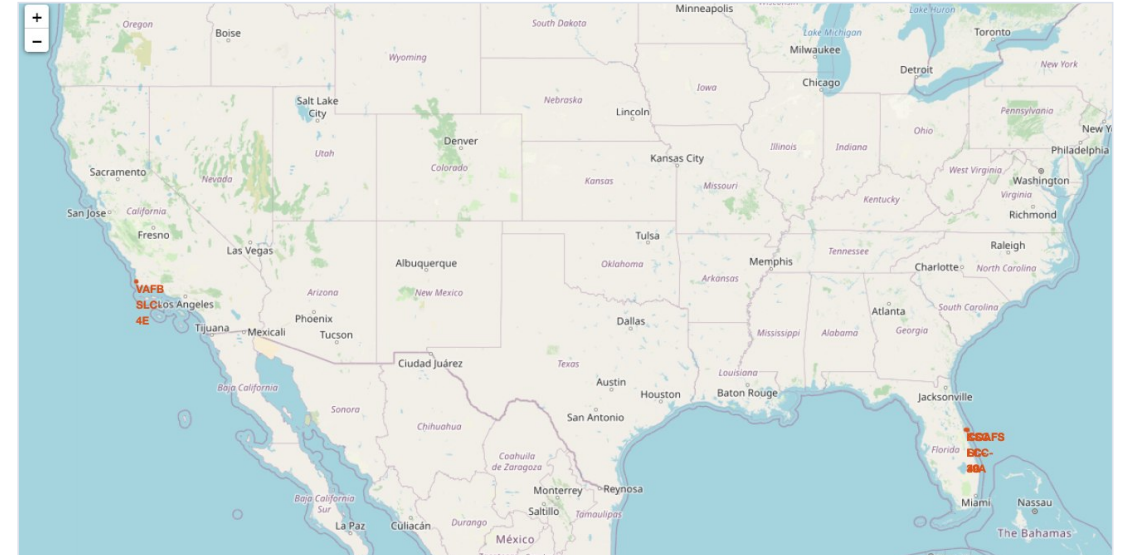
Landing outcome between 2010-06-04 and 2017-03-20	Count
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

Payload query	Result
Max payload mass	15,600 kg
Boosters carrying max payload	F9 B5 B1048.4, B1049.4, B1051.6, B1060.3, B1049.7
Successful drone ship payload filter	Returned successful drone ship landings between 4,000 and 6,000 kg

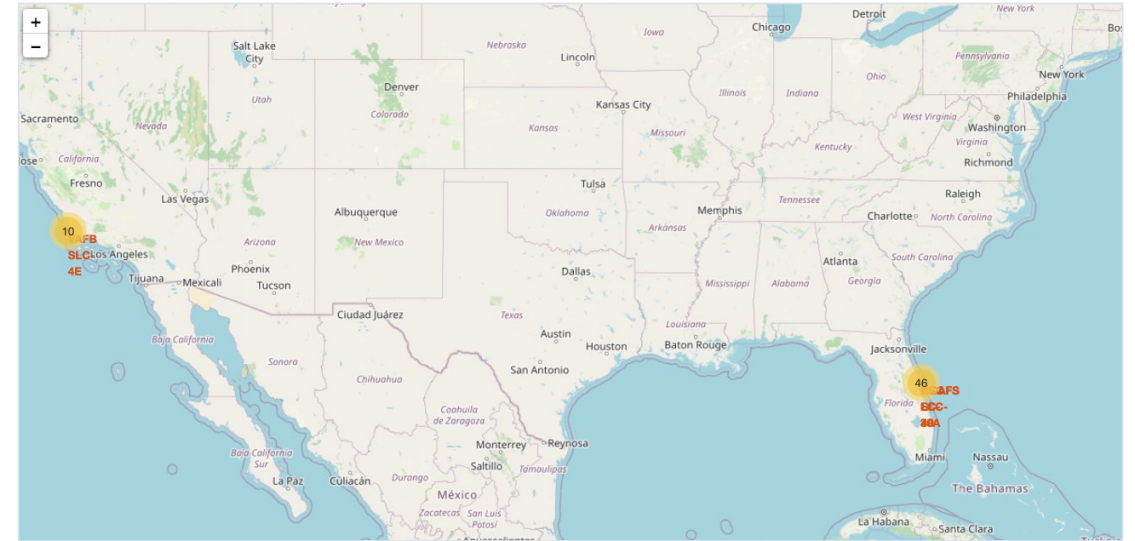
- Outcome counts show that early launch years included a substantial number of no-attempt or failed recovery outcomes.
- Later booster versions carried larger payload masses, which aligns with the broader technology maturity pattern.

Interactive Map Methodology Using Folium

- Mapped all launch site coordinates using Folium.
- Added colored launch outcome markers to compare successful and unsuccessful outcomes by location.
- Used marker clustering to make high-density launch records easier to interpret.
- Added proximity analysis to examine geographic context near coastlines, cities, railways, and highways.

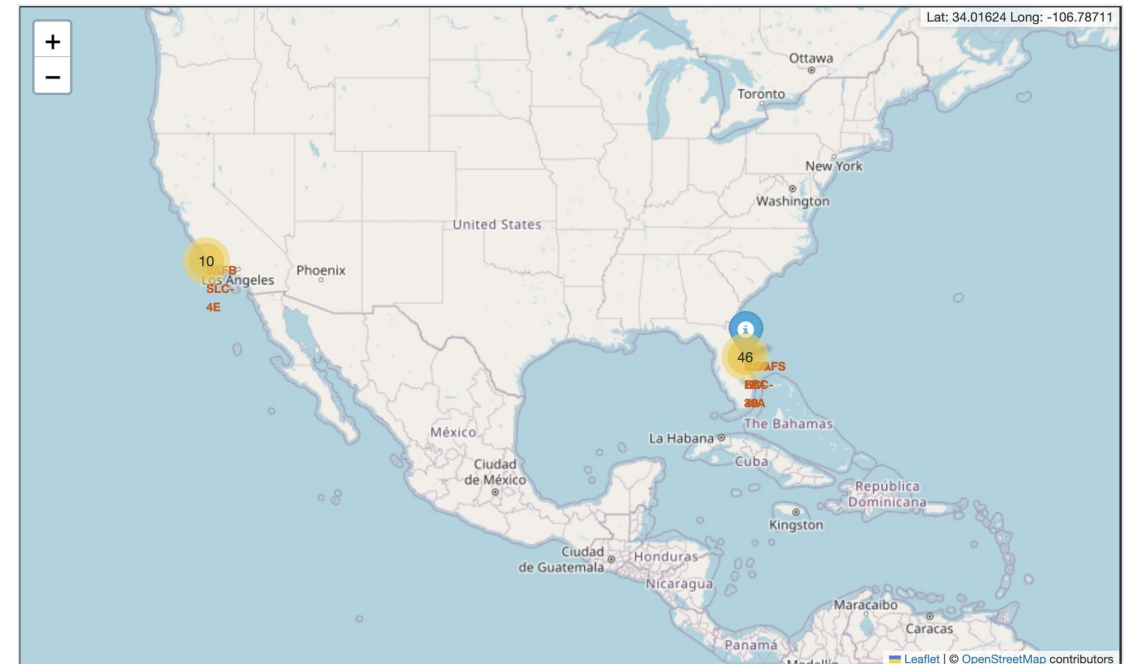
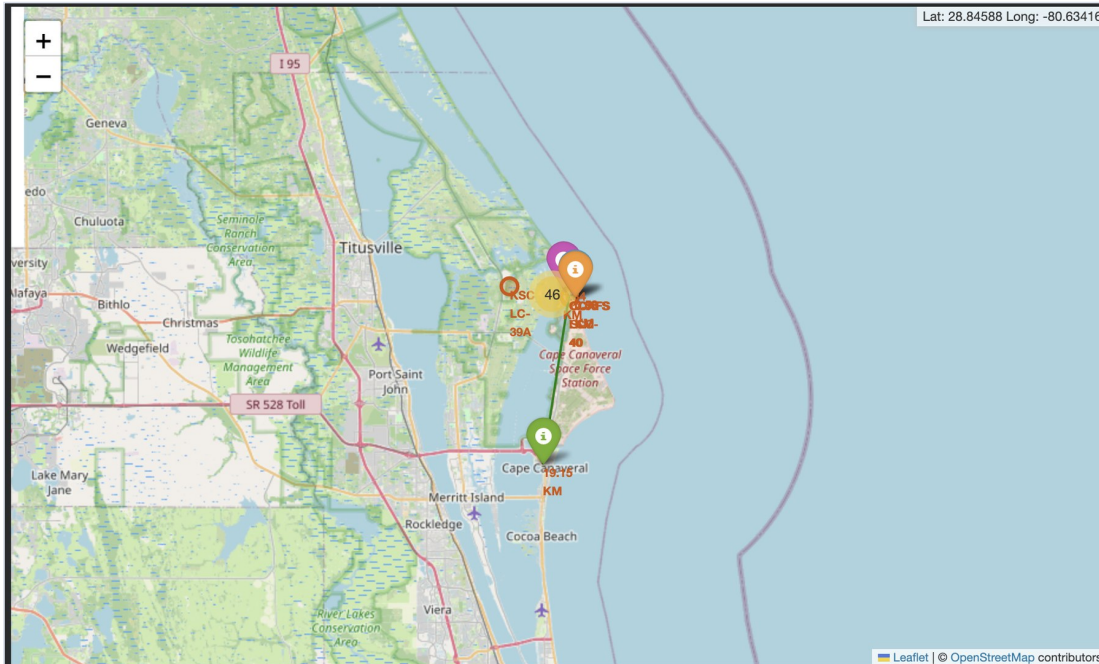


Map-based analysis makes launch geography visible, which is hard to communicate from tables alone.



- Cooper Robinson | GitHub: <https://github.com/cooper-710/Coursera-Capstone>

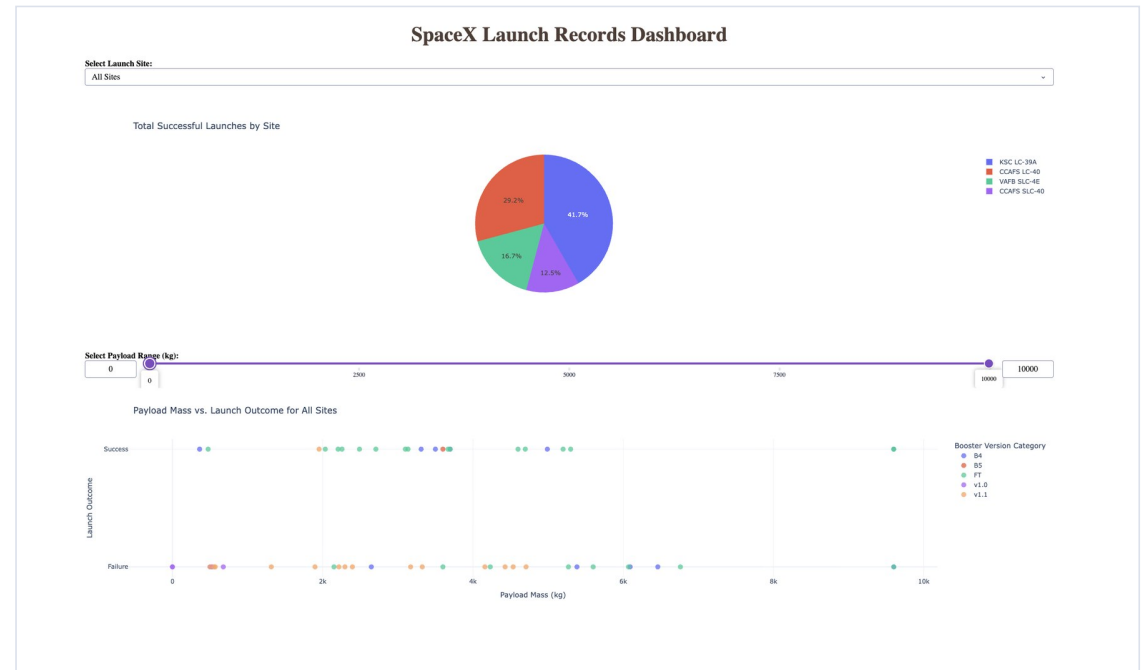
Folium Results: Proximity Analysis



The proximity maps show why launch sites are operationally positioned near open water while still remaining close enough to transportation and support infrastructure.

Plotly Dash Dashboard Methodology

- Built a local Dash app using pandas, Plotly, and Dash callbacks.
- Added a launch site dropdown to switch between all sites and individual launch sites.
- Added a payload range slider to filter launches by payload mass.
- Updated pie and scatter charts interactively based on dropdown and slider inputs.



The dashboard turns static EDA into an interactive tool for exploring success rate by site and payload range.

Dash Results: Successful Launches by Site

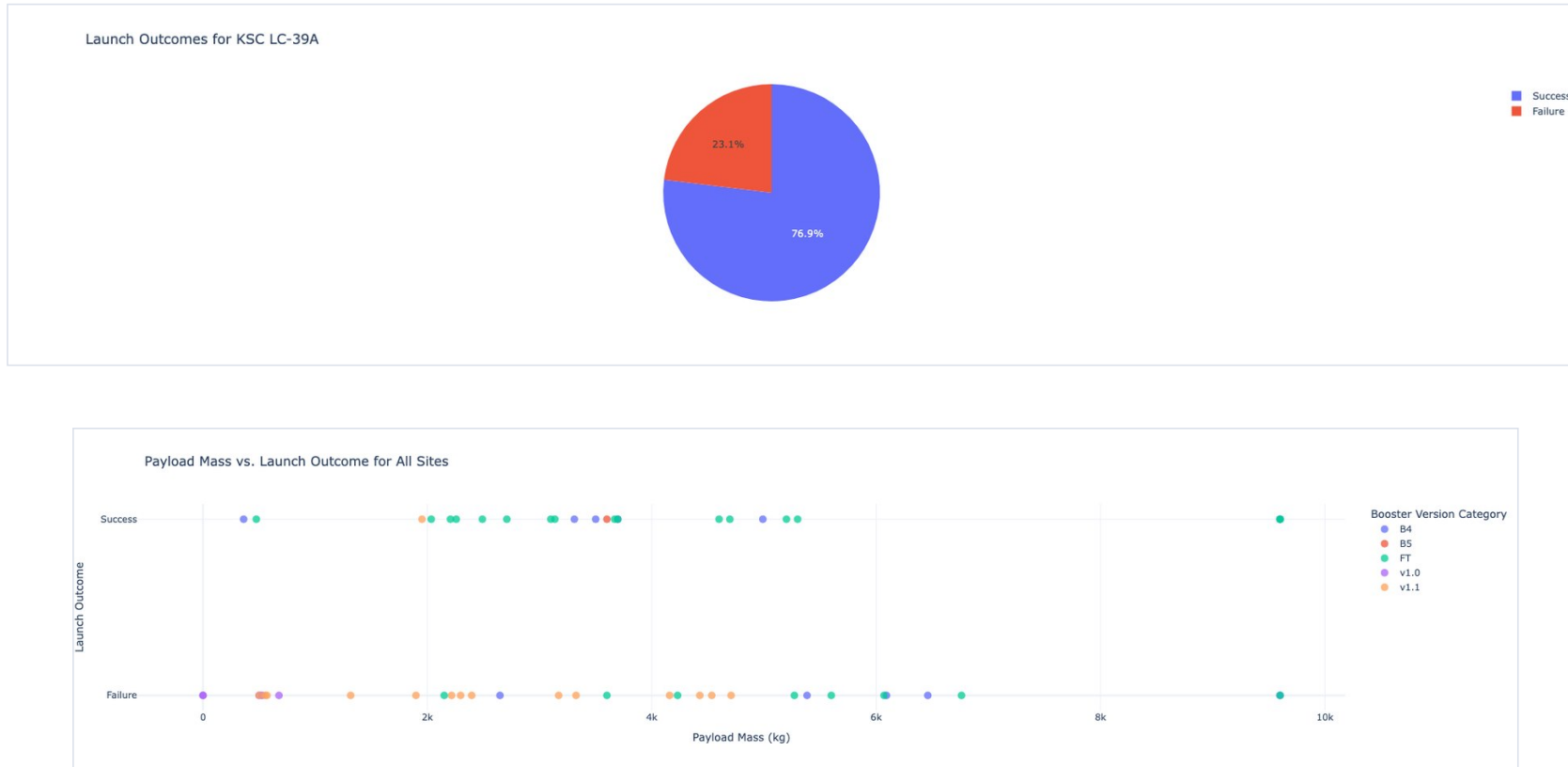
Total Successful Launches by Site



- The all-sites pie chart compares successful launches across major launch sites.
- KSC LC-39A accounts for the largest share of successful launches in the dashboard output.
- This output directly supports the project requirement for Plotly Dash dashboard results.

Dashboard answer: KSC LC-39A appears as the strongest successful-launch contributor in the filtered dashboard view.

Dash Results: Site Selection and Payload Scatter



- The selected-site pie chart compares success versus failure for one launch site.
- The payload scatter shows launch outcome by payload mass and booster version category.
- The scatter plot shows successful landings across a wide payload range, but failures remain present, especially in earlier booster categories.

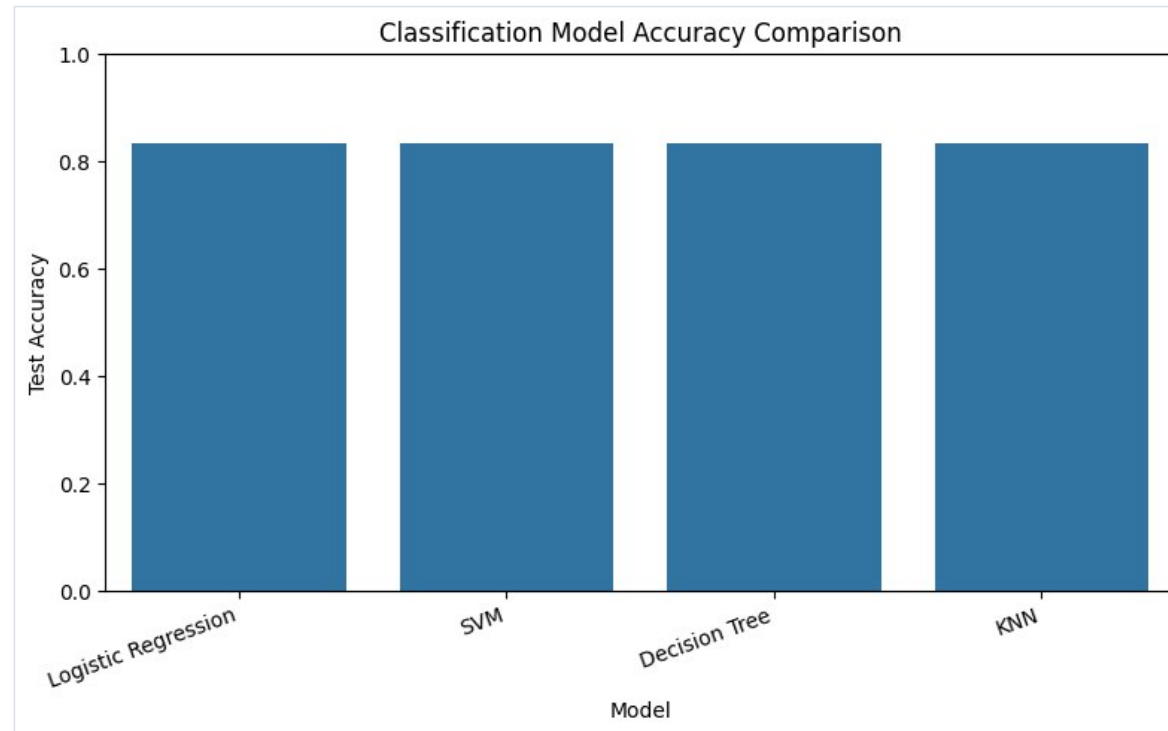
Predictive Analysis Methodology

- Created feature matrix X and target vector Y from the cleaned launch dataset.
- Standardized numerical and encoded categorical features before model training.
- Split data into training and test sets.
- Used GridSearchCV to tune hyperparameters for Logistic Regression, SVM, Decision Tree, and KNN.
- Evaluated each model using test accuracy and confusion matrix output.

Model	Role in comparison
Logistic Regression	Simple linear baseline
SVM	Margin-based classifier
Decision Tree	Interpretable nonlinear model
KNN	Instance-based classifier

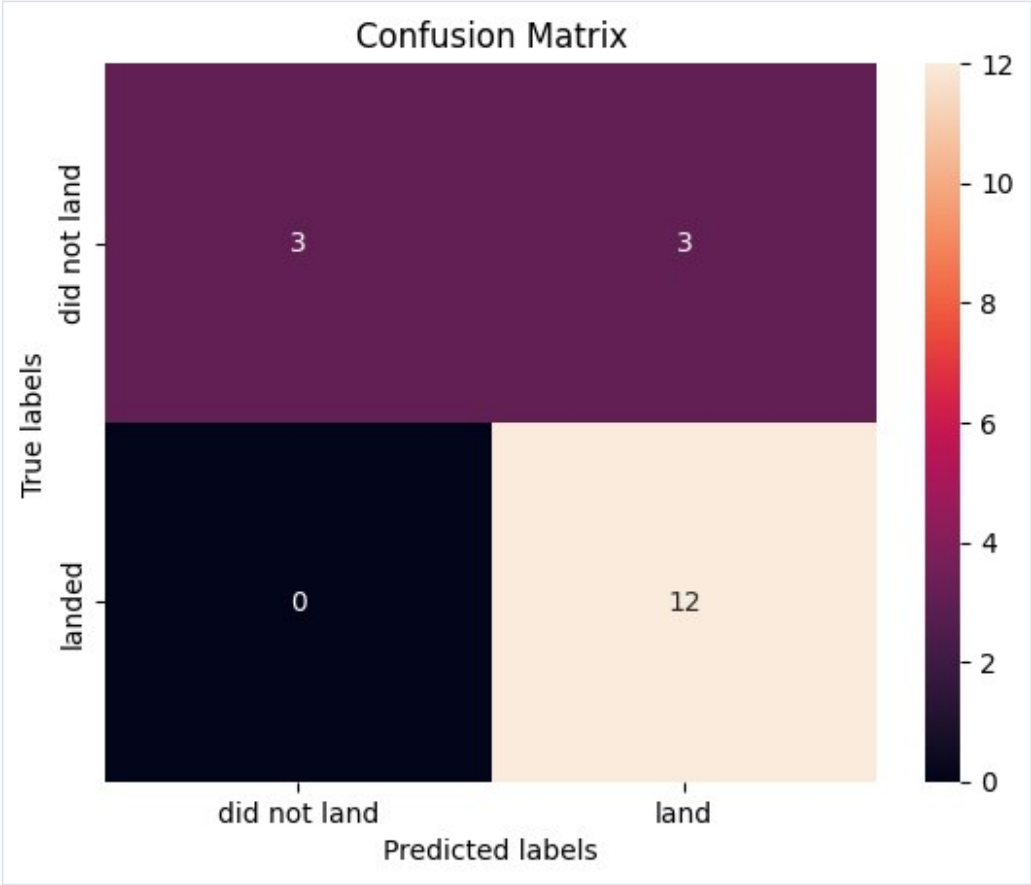
The modeling section focused on comparing several classification approaches rather than relying on one model by default.

Predictive Analysis Results: Model Accuracy



- All four models reached about 83.3% test accuracy.
- Because accuracy was tied, the simplest model, Logistic Regression, is a practical final model choice.
- The similar performance suggests that the dataset is small enough that model choice is less important than feature quality and class balance.

Predictive Analysis Results: Confusion Matrix



Metric	Value
Correct predictions	15 / 18
Test accuracy	83.3%
Correct landed predictions	12
Correct did-not-land predictions	3
False positives	3
False negatives	0

Interpretation: the model predicts successful landings well, but it is weaker at catching failed landing cases.

Key Findings

1

Launch maturity matters

Yearly trend and flight number analysis show success improved over time.

2

Launch context matters

Site, orbit type, and payload mass produce different success patterns.

3

Interactive tools improve analysis

Folium and Dash make site geography and payload filters easier to explore.

4

Prediction is useful but imperfect

Models reached 83.3% accuracy, but failures are harder to classify.

Most important insight: the landing problem is not random. Operational experience and launch configuration provide meaningful predictive signal.

Innovative Insights Beyond the Template

- Failure risk is asymmetric: the final classifier made no false negative landing predictions, but it did label some failed landings as successful. That matters because false confidence is more costly than conservative prediction.
- Payload mass should not be interpreted in isolation. It needs to be read with booster version and launch site because different mission profiles create different risk patterns.
- Launch site comparisons become clearer when static charts are paired with geographic context. Folium adds physical intuition that a regular chart cannot show.
- A future version could turn the Dash app into a mission planning tool with probability output, confidence flags, and model explanations for each prediction.

Creative enhancement used in this report: the project outputs were organized as a complete analysis package with GitHub, notebooks, map screenshots, dashboard screenshots, and model result visuals.

Conclusion

The SpaceX Falcon 9 landing project showed how a full data science workflow can move from raw data to useful operational insight.

- Data collection and wrangling produced a usable supervised learning dataset.
- EDA and SQL identified meaningful patterns in site, orbit, payload, booster, and yearly recovery trends.
- Folium and Dash added interactivity and made the analysis easier to explore.
- Machine learning models predicted landing outcomes with about 83.3% test accuracy.
- The main limitation is failure detection, which should be improved with more launch records, richer booster features, and probability calibration.

Final recommendation: use the model as a decision-support tool, not a standalone decision maker. It is strong enough for pattern detection, but not enough for high-risk launch decisions without additional validation.

Final Deliverable Checklist



Required component	Included
GitHub repository link	Yes
PDF presentation report	Yes
Executive summary and introduction	Yes
Data collection and wrangling methodology	Yes
EDA and SQL results	Yes
Folium map results	Yes
Plotly Dash dashboard results	Yes
Predictive classification results	Yes
Conclusion	Yes
Creativity and innovative insights	Yes

Submission file name: Data Science Capstone Project Report.pdf

Appendix: Additional Visual Outputs



Additional screenshots and notebook outputs are available in the **GitHub images and notebooks folders**.

Repository and Submission Reference

GitHub repository

<https://github.com/cooper-710/Coursera-Capstone>

- Repository includes all completed notebooks, dashboard code, screenshots, README, and requirements file.
- The PDF version of this deck is the file submitted for the final assignment.
- The analysis was completed using the IBM Data Science Capstone SpaceX Falcon 9 project workflow.

File to submit: Data Science Capstone Project Report.pdf